

☑ Problem Statement

Public transportation fleets such as buses and service vehicles frequently face unexpected mechanical failures, resulting in route delays, increased operational costs, and customer dissatisfaction. Most maintenance systems still rely on periodic manual inspections rather than real-time data-driven insights, causing inefficient resource utilization and unplanned breakdowns.

To address this, the objective of this project is to develop a **machine-learning-based predictive maintenance system** that analyzes historical vehicle telemetry and maintenance data to predict component failures in advance. The system will help transportation authorities and fleet managers reduce downtime, optimize maintenance schedules, and improve service reliability.

This study uses the **Vehicle Maintenance Dataset** from Kaggle, which includes sensor readings, vehicle performance metrics, component health data, and maintenance logs. The dataset enables training models to forecast potential mechanical failures and identify critical warning parameters, contributing to data-driven decision-making in transportation maintenance.

Additionally, the project will include a **web-based dashboard/website showcase** where model predictions, insights, and vehicle health status will be displayed, enabling interactive visualization for fleet managers.

☑ Dataset Used

Vehicle Maintenance Dataset (Kaggle)

🔗 <https://www.kaggle.com/datasets/chavindudulaj/vehicle-maintenance-data>

🎯 Expected Outcome

- Early prediction of vehicle breakdown risk
 - Reduced maintenance cost & downtime
 - Improved fleet performance and operational efficiency
 - Interactive dashboard for real-time monitoring & alerts
 - **Website showcase for project demonstration**
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🔧 Tools & Tech Stack

Category	Tools Used
Programming Language	Python
Data Handling & Analysis	Pandas, NumPy
Modeling & Machine Learning	Scikit-Learn, XGBoost, Random Forest
Visualization	Matplotlib, Seaborn, Power BI / Tableau (optional)
Dashboard / Web Deployment	Streamlit / Flask / Django
Data Source	Kaggle Vehicle Maintenance Dataset
Version Control	Git, GitHub
Development Environment	Jupyter Notebook / VS Code
Model Serialization	Pickle / Joblib
